1 Inverse Trigonometry and Identities

Inverse trigonometric functions recover principal angles from trigonometric values. Their definitions require restrictions of the original domains. Trigonometric identities, in turn, express relationships among functions of the same angle and provide methods for simplifying and evaluating expressions.

Learning objectives. Determine the domains and ranges of the principal inverse trigonometric functions; evaluate and compose inverse and trigonometric functions; recover exact values from right triangles; prove identities using fundamental relations; and derive and apply angle-addition and double-angle formulas.

1.1 Inverse functions and principal branches

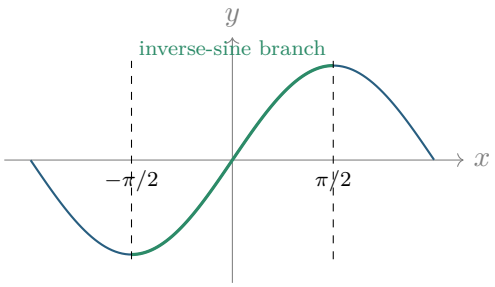
The notation $\sin^{-1} x$ commonly denotes $\arcsin x$, the inverse sine function. It must not be confused with $\csc x = 1/\sin x$. The inverse returns an angle from a prescribed trigonometric value, whereas the reciprocal returns a multiplicative inverse of that value.

A periodic function is not one-to-one on its full domain and therefore does not admit a global inverse function. To define the principal inverse sine, sine is restricted to $[-\pi/2, \pi/2]$, where it is strictly increasing and maps onto $[-1, 1]$. Similarly, cosine is restricted to $[0, \pi]$ and tangent to $(-\pi/2, \pi/2)$.

$f(x)$	Domain	Range
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$

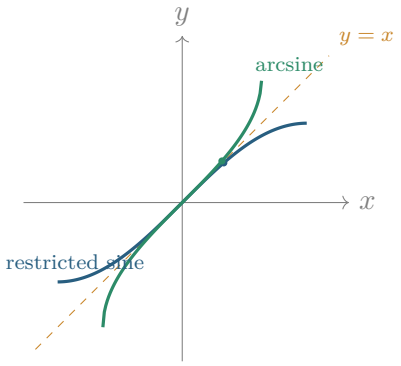
Example 1.1. $\arcsin(1/2) = \pi/6$, $\arccos(-1/2) = 2\pi/3$, and $\arctan(-1) = -\pi/4$.

Example 1.2. $\arcsin(\sin(5\pi/6)) = \pi/6$, not $5\pi/6$, because the inverse reports its principal value.



1.2 Reflection across the diagonal

The graphs of an invertible function and its inverse are reflections across the line $y = x$: if (a, b) lies on the graph of f, then (b, a) lies on the graph of f^{-1}. For inverse sine, only the restricted sine branch is reflected, rather than the entire periodic sine graph.



The reflected domains and ranges exchange roles. In particular, $\arcsin : [-1,1] \to [-\pi/2, \pi/2]$ whereas the restricted sine maps $[-\pi/2, \pi/2]$ onto $[-1,1]$.

Try it. List three points on $y = \cos x$ for $0 \le x \le \pi$, and interchange their coordinates to obtain points on $y = \arccos x$.

1.3 Compositions and triangle geometry

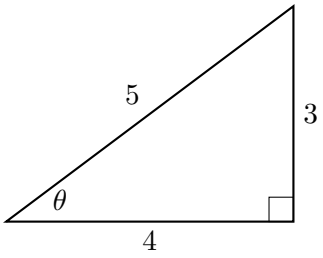
The identity $\sin(\arcsin x) = x$ holds for $-1 \le x \le 1$, but $\arcsin(\sin\theta) = \theta$ only for θ in the principal range. Suppose $\theta = \arctan(3/4)$. Then a right triangle has legs $3, 4$ and hypotenuse 5, so

$$\sin(\arctan(3/4)) = \frac{3}{5}.$$

For the inverse compositions, the domain restrictions determine which cancellations are valid:

$$\arcsin(\sin x) = x \quad \text{for } -\pi/2 \le x \le \pi/2, \qquad \sin(\arcsin u) = u \quad \text{for } -1 \le u \le 1.$$

The first identity does not hold for arbitrary real x, since the arcsine output is restricted to its principal range. Analogous statements hold for cosine and tangent on their chosen branches.



$\theta = \arctan(3/4)$

$\sin\theta = 3/5$

Example 1.3. To evaluate $\cos(\arcsin(-5/13))$, let $\theta = \arcsin(-5/13)$. Since $\theta \in [-\pi/2, \pi/2]$, its cosine is nonnegative. Thus

$$\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}.$$

The principal range determines the sign of the square root.

Try it. Evaluate $\sin(\arccos(-3/5))$ exactly. Explain the sign of the result using the principal range of arccosine.

1.4 Identities and their domains

From the unit-circle equation we obtain

$$\boxed{\sin^2 x + \cos^2 x = 1}.$$

Dividing by $\cos^2 x$ (only where $\cos x \neq 0$) gives $1+\tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$ gives $1+\cot^2 x = \csc^2 x$ where $\sin x \neq 0$.

Example 1.4. For $\sin x \neq 0$,

$$\frac{1-\cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x.$$

This is an algebraic proof on the shared domain, not a numerical check at a few inputs.

1.5 Symmetry identities

The unit circle is symmetric about both coordinate axes. Reflection across the horizontal axis changes the sign of the vertical coordinate without changing the horizontal coordinate. Consequently,

$$\cos(-x) = \cos x, \qquad \sin(-x) = -\sin x.$$

The cosine function is *even*, while sine is *odd*. Taking quotients gives $\tan(-x) = -\tan x$ wherever tangent is defined.

A half-turn changes both coordinate signs:

$$\cos(x+\pi) = -\cos x, \qquad \sin(x+\pi) = -\sin x.$$

These relations explain the π-periodicity of tangent.

Try it. Use the unit-circle symmetries to evaluate $\sin(-7\pi/6)$ and $\cos(-7\pi/6)$ without a calculator.

1.6 Angle addition and double angles

Two successive plane rotations produce the sum identities

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Set $\beta = -\gamma$ for difference identities and $\beta = \alpha$ for double-angle identities:

$$\sin(2x) = 2\sin x\cos x, \qquad \cos(2x) = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x.$$

Example 1.5. $\sin(75^\circ) = \sin(45^\circ + 30^\circ) = (\sqrt{6}+\sqrt{2})/4$.

1.7 A geometric derivation of the addition formulas

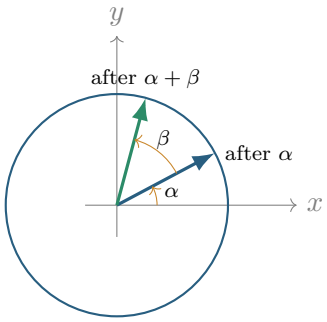
Rotation through an angle α sends a point (x, y) to

$$\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

In particular, apply this rotation to the unit-circle point $(\cos\beta, \sin\beta)$. The result must be the point $(\cos(\alpha+\beta), \sin(\alpha+\beta))$, since successive rotations add their angles. Matrix multiplication gives

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Thus the sum formulas express the geometry of composing rotations, rather than independent algebraic rules.



1.8 Double-angle and half-angle consequences

Setting $\alpha = \beta = x$ gives

$$\sin 2x = 2\sin x \cos x, \qquad \cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1.$$

Rearranging the cosine identity produces the power-reduction relations

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

These expressions are particularly useful when simplifying squared trigonometric functions.

Example 1.6. The exact value of $\cos 15^\circ$ follows by using $15^\circ = 45^\circ - 30^\circ$:

$$\cos 15^\circ = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

Try it. Use the addition identities to find $\sin 105^\circ$ exactly. Then verify the sign from the quadrant containing 105°.

1.9 Applications: measured angles

An observation of a tower from 40 m away gives $\tan\theta = 3/4$. The elevation angle is $\arctan(3/4) \approx 36.9^\circ$, while the height is exactly $40(3/4) = 30$ m. Inverse functions recover angles; ratios can often recover distances directly.

1.10 Recovering angles in measurement problems

The inverse function identifies an angle from a ratio, provided the physical geometry determines an appropriate branch. If a right triangle has adjacent side 40 and opposite side 30, its acute angle satisfies

$$\theta = \arctan(30/40) = \arctan(3/4) \approx 36.87^\circ.$$

The same geometry gives the opposite side directly from $h = 40\tan\theta$, when the angle is already known. In a measurement context, the acute-angle restriction is supplied by the geometry of the triangle.

1.11 Exercises

1. Evaluate exactly $\arcsin(-\sqrt{2}/2)$ and $\arctan(\sqrt{3})$.

2. Explain why $\arcsin(\sin(3\pi/4)) \neq 3\pi/4$.

3. Find $\cos(\arcsin(5/13))$ with a triangle.
4. Find $\sin(\arctan(-2))$ exactly, including its sign.
5. Prove $(\sec^2 x - 1)/\tan x = \tan x$ on its domain.
6. Prove $(\sin x + \cos x)^2 = 1 + \sin(2x)$.
7. Calculate $\cos(75^\circ)$ exactly by angle addition.
8. Express $\cos(2x)$ using $\sin x$ alone.
9. State the domain and range of $\arccos x$, and explain why the restriction $[0, \pi]$ is appropriate.
10. Evaluate $\arccos(\cos(5\pi/3))$ exactly.
11. Find $\tan(\arccos(4/5))$ exactly using a right triangle.
12. Prove $\sin(-x) = -\sin x$ geometrically and use it to compute $\sin(-\pi/3)$.
13. Verify the cosine difference identity by replacing β with $-\beta$ in the cosine sum identity.
14. Prove $\cos^2 x = (1 + \cos 2x)/2$ on all of $\mathbb{R}$.
15. Find $\sin 15^\circ$ exactly and state its sign.
16. Explain why $\arctan(\tan(3\pi/4))$ is not $3\pi/4$.